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Student Number

2014

**TRIAL HIGHER SCHOOL CERTIFICATE
EXAMINATION**

Mathematics

**** July 2014**

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using blue or black pen
Black pen is preferred
- Approved calculators may be used
- A table of standard integrals is provided at the back of this paper
- In Questions 11-16 show relevant mathematical reasoning and/or calculations
- Start a new booklet for each question

Total Marks – 100

Section I - Pages 2 - 6

10 marks

- Attempt Questions 1 – 10
- Allow about 15 minutes for this section

Section II - Pages 6 - 12

90 marks

- Attempt Questions 11 – 16
- Allow about 2 hour and 45 minutes for this section

Question	Marks
1 - 10	/10
11	/15
12	/15
13	/15
14	/15
15	/15
16	/15

THIS QUESTION PAPER MUST NOT BE REMOVED FROM THE EXAMINATION ROOM

This assessment task constitutes 40% of the Higher School Certificate Course Assessment

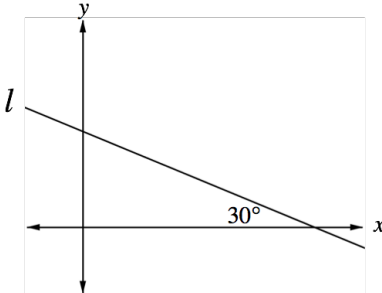
Section I

10 marks

Attempt Question 1 – 10

Allow about 15 minutes for this section

Use the multiple – choice answer sheet for Questions 1 – 10

1	If $\frac{1}{2+\sqrt{3}} = a + b\sqrt{3}$ then (A) $a = 2$ and $b = 1$ (B) $a = 2$ and $b = -1$ (C) $a = -2$ and $b = -1$ (D) $a = 4$ and $b = 1$
2	Which of the following is true for the equation $7x^2 - 5x + 2 = 0$ (A) No real roots (B) One real root (C) Two real roots (D) Three real roots
3	The diagram shows the line l.  What is the slope of the line l? (A) $\sqrt{3}$ (B) $-\sqrt{3}$ (C) $\frac{\sqrt{3}}{3}$ (D) $-\frac{\sqrt{3}}{3}$

4	What is the solution to the inequality $2x + 1 \leq 3$ (A) $x \geq 1$ or $x \geq -2$ (B) $x \geq 1$ or $x \geq 2$ (C) $x \leq 1$ or $x \geq -2$ (D) $x \leq 1$ or $x \geq 2$
5	What is the derivative of $\frac{x}{\sin x}$? (A) $\frac{\sin x + x \cos x}{\sin^2 x}$ (B) $\frac{\sin x - x \cos x}{\sin^2 x}$ (C) $\frac{x \sin x - \cos x}{\sin^2 x}$ (D) $\frac{x \cos x - \sin x}{\sin^2 x}$
6	The rule of the function whose graph is given above is: (A) $y = x - 2 + 2$ (B) $y = x + 2 - 2$ (C) $y = 2 - x - 2$ (D) $y = 2 + x + 2$

7	A bag contains the numbers 1 – 20. A single number is drawn. What is the probability that it is a multiple of 2 OR a multiple of 5? (A) $\frac{3}{5}$ (B) $\frac{9}{10}$ (C) $\frac{1}{2}$ (D) $\frac{8}{25}$
8	The domain of the function $y = \log_e(2x - 1)$ is: (E) all real x , $x \neq \frac{1}{2}$ (F) all real x , $x \geq \frac{1}{2}$ (G) all real x , $x \leq \frac{1}{2}$ (H) all real x , $x > \frac{1}{2}$

9	Water is being poured into a long cylindrical storage tank of radius 2 metres, with its circular base on the ground, at a rate of 2 cubic metres per second. A diagram of a cylindrical storage tank. The radius of the top circular face is labeled as 2 m. The tank is partially filled with water, and the depth of the water is indicated by a vertical double-headed arrow labeled 'depth'. (A) $\frac{1}{8\pi}$ (B) $\frac{1}{4\pi}$ (C) $\frac{1}{2\pi}$ (D) 2π
10	Which of the following is a graph of $y = -2\cos\left(2x - \frac{\pi}{3}\right)$
(A)	(B)
(C)	(D)

Section II

90 marks

Attempt Questions 11 – 16

Allow about 2 hours and 45 minutes for this section

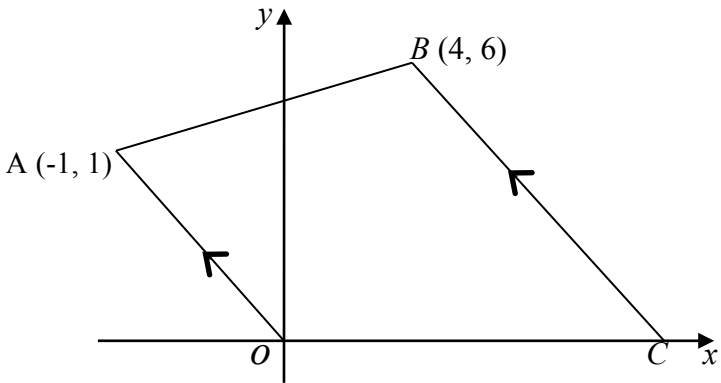
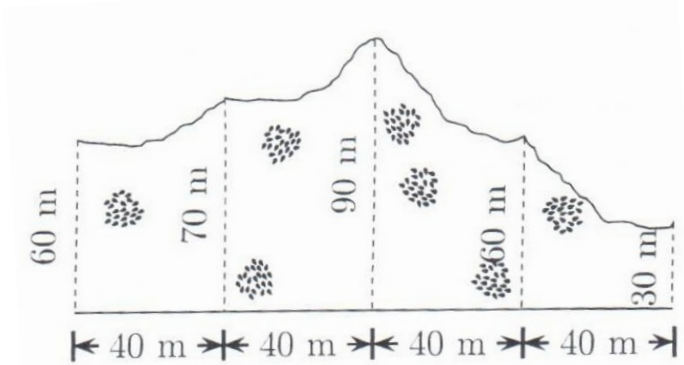
Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

In Questions 11 – 14, your responses should include relevant mathematical reasoning and/or calculations.

Question 11(15 marks) Use a SEPARATE writing booklet

(a)	Consider the line $x + 2y - 4 = 0$ Sketch this line showing clearly the intercepts on both axes.	2
(b)	Solve $\frac{5}{2a} - \frac{3}{2a} = \frac{1}{2}$	2
(c)	Solve $ x - 6 = 11$	2
(d)	Find the equation of the tangent to the curve $x^2 = 4y$ at the point (6, 9)	2
(e)	Solve for θ where $0 \leq \theta \leq \pi$, $3 \tan^2 \theta = 1$	2
(f)	Solve $\log_2 7 = x$ giving your answer to 3 significant figures	1
(g)	Differentiate (i) $4x \sin \frac{x}{4}$ (ii) $\frac{4xe^x + 3x^2}{x}$	2 2

Question 12(15 marks) Use a SEPARATE writing booklet

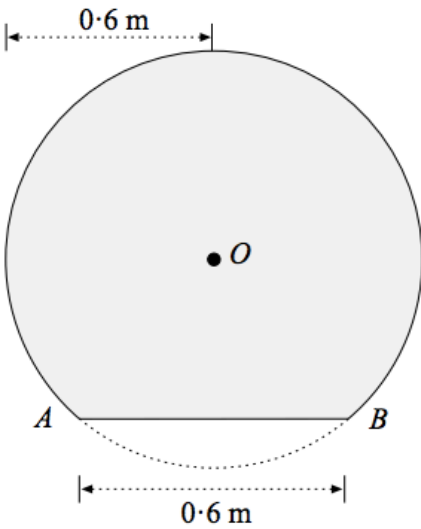
(a)	(iii) Find $\int \pi dx$. (iv) Find $\int \frac{6}{3x+1} dx$. (v) Find $\int_0^1 (\sqrt{x} - \sqrt[3]{x}) dx$	1 2 2
(b)	In the diagram, $OABC$ is a trapezium with $OA \parallel CB$. The coordinates of O, A and B. The coordinates of O, A, and B are $(0, 0)$, $(-1, 1)$ and $(4, 6)$ respectively.  (i) Calculate the distance of OA. (ii) Find the equation of the line BC, and hence find the coordinates of C. (iii) Show that the perpendicular distance from O to the line BC is $5\sqrt{2}$.	1 2 2
(c)	Shade the region in the plane defined by $y < 3$ and $y \geq x^2 - 4x + 3$.	2
(d)	The diagram shows the land that Jane bought near a lake.  (i) Use the trapezoidal rule with 5 function values to find an estimate for the area of this land (ii) Is the area obtained in the previous part smaller or larger than the exact area? Give reasons for your answer.	2 1
	End of Question 12	

Question 13(15 marks) Use a SEPARATE writing booklet

(a)	Find the values of a, b c for which $a + b(x + 2) + cx(x + 2) \equiv 6x^2 + x - 2$	3
(b)	NOT TO SCALE In the diagram above, WXYZ is a rectangle with WX=12cm and WZ=9cm. WP is perpendicular to XZ. Copy the diagram onto your worksheet (i) Find the length of XZ (ii) Prove that $\triangle WXP$ is similar to $\triangle ZXW$. (iii) Hence find the length of XP.	1 3 1
	Question 13 continues on page 9	

	Question 13 (continued)	
(c)	On a set of coordinate axes, A is the point $(-3, 4)$ and B is the point $(4, -2)$. i.) Show that the equation of the line AB is $6x + 7y - 10 = 0$ ii) What is the equation of the line parallel to AB through the point C $(-2, 0)$ iii) What is the shortest distance from C to the line AB. Give your answer in exact form. iv) Write down the equation of the circle which has a centre C and AB as a tangent.	2 2 2 2
	End of Question 13	

Question 14(15 marks) Use a SEPARATE writing booklet

(a)	Toby has an old laptop that causes an error when he turns it on 15% of the time. He takes it to a repair technician who asks him to turn the computer on and off to demonstrate the error. (i) What is the probability the computer doesn't demonstrate the error on the first three attempts to start it? (ii) How many times must Toby restart the computer to have a greater than 75% chance of demonstrating the error at least once to the technician?	1 2
(b)	 A table top is in the shape of a circle with a small segment removed as shown. The circle has a centre O and radius 0.6 metres. The length of the straight edge AB is also 0.6 metres. (i) Explain why $\angle AOB = \frac{\pi}{3}$ (ii) Find the area of the table-top.	2 2

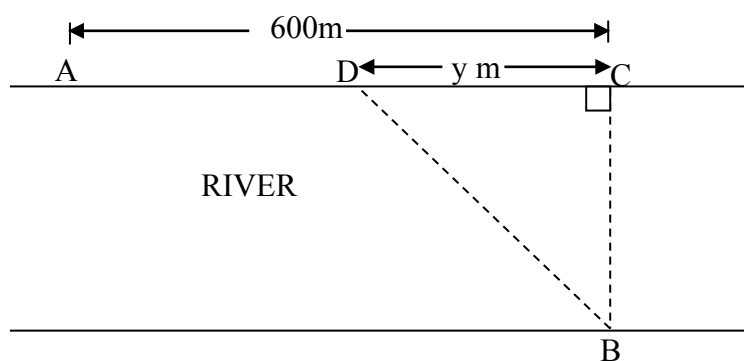
(c)	Find the exact area bounded by $y = \sin(2x)$ and the x -axis between $x = 0$ and $x = \pi$	3
(d)	A new species of super ant from the Black Sea (<i>Lasius Neglectus</i>) is spreading throughout the world's ports. It rapidly supplants local ant populations and takes over, hosting 10 to 100 times the number of worker ants as local species. New queens often stay with colony making yet more workers. A small colony was found in a backyard in Sydney, estimated at only 10 queens. After 3 years the nest covers an entire suburb involving an estimated 800 queens. Their growth takes the form $P = P_0 e^{kt}$ where k is a positive constant and P_0 is the original population discovered at time $t = 0$ years. (i) Find the value of k. (ii) The largest colony ever found had 35000 queens. How long until the Sydney colony reaches that many queens? (iii) What could limit the accuracy of this formula for large values of t?	2 2 1
End of Question 14		

Question 15(15 marks) Use a SEPARATE writing booklet

(d)	A point P moves in a straight line, so that its displacement from the origin after t seconds is given by: $x = 3e^{-5t} + 3t - 2$ (i) Find the initial displacement, velocity and acceleration of the particle. 3 (ii) Find the time at which the particle is stationary and hence find its displacement and acceleration at that time. 3 (iii) Find an expression for the acceleration of the particle in terms of its velocity. 2 (iv) Describe the motion of particle P as $t \rightarrow \infty$ 1	
(e)	The equation below refers to the filtering cycle of a pump in Helen's garden. The flow rate of the volume of water that the filter pumps water into and out of a pond in litres per minute, is given by $\frac{dV}{dt} = 20 \sin \frac{\pi}{35} t .$ (i) If the pump started at 8.55pm, what is the first time after 8.55 pm at which the flow rate is zero? 2 (ii) If the pond is initially empty find an expression for the volume, V, of water in the pond after t minutes. 2 (iii) Find the maximum volume of water in the pond during the filtering cycle. Leave your answer in terms of π. 2	
	End of Question 14	

Question 16 (15 marks) Use a SEPARATE writing booklet

(f)



Jackson's farm house, at 'A', is on the edge of a river 200 metres wide. His horse stables, at 'B', is on the other side of the river. Jackson wants to run an electricity cable from his farm house to the stables.

It costs \$1,750 per 100m to lay cables along the river's edge and \$2,900 to lay the cable under water.

'C' is the point on the river closest to 'B' and the distance AC is 600

The point 'D' on the rivers bank is at a distance of 'y' metres from 'C'.

- (i) Find the total cost in laying the cable in a straight line from A to C and then in a straight line from C to B.
- (ii) What is the cost of running the cable directly from A to B.
- (iii) Let \$F be the total cost of laying the cable in a straight line from A to D, and then in a straight line from D to B.

Show that $F = 1750 (6 - 0.01y) + 290 (\sqrt{400 + 0.01y^2})$

- (iv) Find the minimum cost of laying the cable.

- (v) A week before Jackson accepted the minimum costs as determined in part (iii), new technology revealed that the cost of laying the cable underwater was reduced to \$2000 per 100 metres. In this case determine the path for laying the cable in order to minimise the costs.

(g)	The average monthly temperature, $T^{\circ}\text{C}$, in Canberra can be modelled by the formula $T = 7 \sin(nx + 1.5) + 13$, where n = a constant value and x = the number of the month of the year (that is, January =1, February =2....) (i) According to the model, what are the maximum and minimum average monthly temperatures in Canberra? (ii) The period of the function is 12. Determine the value of n correct to 2 decimal places. (iii) Which month has the lowest average monthly temperature?	2 1 2

Yr 12 Mathematics Trial 2014

Answers.

1 B

2 A

3 D

4 C

5 B

6 B

7 A

8 D

9 C

10 B

QUES 11

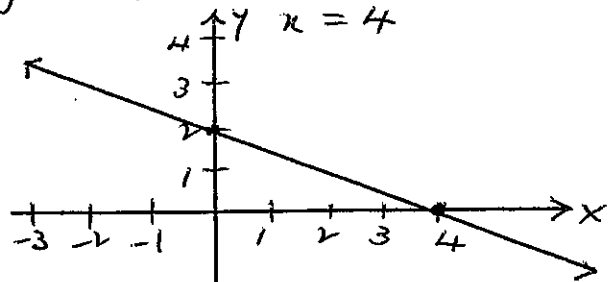
$$a) x + 2y - 4 = 0$$

$$x = 0, 2y - 4 = 0$$

$$y = 2$$

$$y = 0, x - 4 = 0$$

$$x = 4$$



b) see below

$$c) |x - 6| = 11$$

$$x - 6 = 11 \text{ or } x - 6 = -11$$

$$x = 17 \text{ or } x = -5$$

$$d) x^2 = 4y \quad (6, 9)$$

$$y = \frac{x^2}{4}$$

$$y' = \frac{2x}{4} = \frac{x}{2}$$

$$f'(6) = \frac{6}{2} = 3$$

$$\text{equation, } y - y_1 = m(x - x_1)$$

$$y - 9 = 3(x - 6)$$

$$y - 9 = 3x - 18$$

$$0 = 3x - y - 9$$

$$e) \frac{5}{2a} - \frac{3}{2a} = \frac{1}{2a}$$

$$\frac{2}{2a} = \frac{1}{a}$$

$$a = 2$$

① mark for showing intercepts algebraically

① mark for sketch of decent size, clearly showing intercepts.

* Question well done.

① mark for each solution = 2 marks

* Question generally well done

① mark for correct derivative / value for gradient

① mark for equation of tangent

* Some students still not taking derivative and substituting correctly.

① mark for simplification

① mark for solution

* well done.

$$d) 3 \tan^2 \theta = 1 \quad 0 \leq \theta \leq \pi$$

$$\tan^2 \theta = \frac{1}{3}$$

$$\tan \theta = \pm \frac{1}{\sqrt{3}}$$

for $0 \leq \theta \leq \pi$, θ is in 1st, 2nd quadrants only.

$$\therefore \text{basic/acute } \theta = \frac{\pi}{6}$$

$$\therefore \theta = \frac{\pi}{6}, \pi - \frac{\pi}{6}$$

$$= \frac{\pi}{6}, \frac{5\pi}{6}$$

① mark for $\pm \frac{1}{\sqrt{3}}$

① mark for correct solutions.

* many students lost negative sign i.e. only gave $\tan \theta = \frac{1}{\sqrt{3}}$

* some students did not use radians

* some students used $0 < \theta \leq 2\pi$

$$f) \log_2 T = x$$

$$x = \frac{\log T}{\log 2}$$

$$= 2.807354922$$

$$= 2.81 \text{ (3 s.f.)}$$

① mark for correct value, with no penalty for significant figures.

* students need to check/know difference between s.f. and decimal places.

$$g) i) \frac{d}{dx} 4x \sin \frac{x}{4}$$

(PRODUCT RULE)

$$= VU' + UV'$$

$$= \sin \frac{x}{4} \cdot 4 + 4x \cdot \frac{1}{4} \cos \frac{x}{4}$$

$$= 4 \sin \frac{x}{4} + x \cos \frac{x}{4}$$

① mark for each part i.e. VU' , UV' = 2 marks.

* several students cancelled incorrectly i.e. $(\sin \frac{x}{4}) \times 4 \neq \sin x$.

* also $\frac{d}{dx} \sin \frac{x}{4} = \frac{1}{4} \cos \frac{x}{4}$ not $4 \cos \frac{x}{4}$

$$ii) \frac{d}{dx} \frac{4xe^x + 3x^2}{x}$$

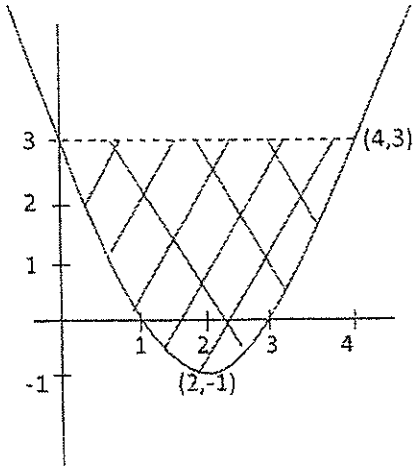
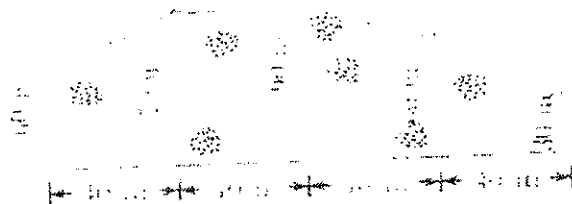
$$= \frac{d}{dx} 4e^x + 3x$$

$$= 4e^x + 3 \text{ ①}$$

*

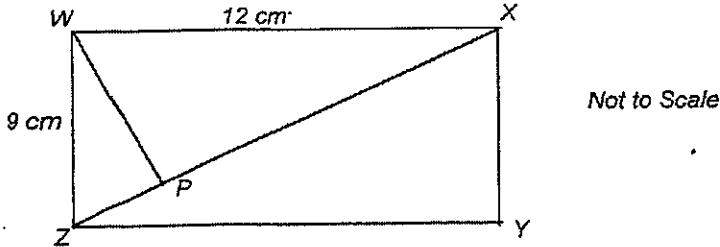
① mark each part, although most students made this question complicated by using the quotient rule.

Question 12	Answer	Mark
a. i)	$\int \pi dx = \pi x + C$	① mark to integrate the equation
ii)	$\int \frac{6}{3x+1} dx = 2 \int \frac{3}{3x+1} dx$ $= 2 \ln(3x+1) + C$	① mark for working. ① mark for correct answer.
iii)	$\int_0^1 (\sqrt{x} - \sqrt[3]{x}) dx = \left[\frac{2}{3} x^{\frac{3}{2}} - \frac{3}{4} x^{\frac{4}{3}} \right]_0^1$ $= \left(\frac{2}{3} (1)^{\frac{3}{2}} - \frac{3}{4} (1)^{\frac{4}{3}} \right) - \left(\frac{2}{3} (0)^{\frac{3}{2}} - \frac{3}{4} (0)^{\frac{4}{3}} \right)$ $= \left(\frac{2}{3} - \frac{3}{4} \right) - (0 - 0)$ $= -\frac{1}{12}$	① mark for working. ① mark for correct answer.
b. i)	$OA = \sqrt{(0+1)^2 + (0-1)^2}$ $= \sqrt{2} \text{ units}$	① mark for working with correct answer
ii)	$BC \parallel AO$ $\therefore m_{(BC)} = m_{(AO)} = -1$ Equation of the line BC: $y - y_1 = m(x - x_1)$, given $B(4, 6)$ $y - 6 = -1(x - 4)$ $y - 6 = -x + 4$ $x + y - 10 = 0$. When $y = 0$, $x = 10$. $\therefore C(10, 0)$.	① mark for finding equation of the line ① mark for finding point $C(10, 0)$.
iii)	$d = \frac{ ax_1 + by_1 + c }{\sqrt{a^2 + b^2}}$ The equation of BC is $x + y - 10 = 0$ and (x_1, y_1) is $O(0, 0)$. $\therefore d = \frac{ 1 \times 0 + 1 \times 0 - 10 }{\sqrt{1^2 + 1^2}}$ $= \frac{10}{\sqrt{2}}$ $= \frac{10}{\sqrt{2}} \times \frac{\sqrt{2}}{\sqrt{2}}$ $= 5\sqrt{2} \text{ units}$	① mark for applying perpendicular distance formula ① mark for correct answer.

c.		① mark for graph ① mark for region																		
d. i)	We have the table of values <table border="1" data-bbox="453 934 1034 1046"> <tr> <td>x</td> <td>0</td> <td>40</td> <td>80</td> <td>120</td> <td>160</td> </tr> <tr> <td>y</td> <td>60</td> <td>70</td> <td>90</td> <td>60</td> <td>30</td> </tr> <tr> <td></td> <td>y_0</td> <td>y_1</td> <td>y_2</td> <td>y_3</td> <td>y_4</td> </tr> </table> Trapezoidal rule: $A \approx \frac{h}{2} (y_0 + 2(y_1 + y_2 + y_3) + y_4)$ $A \approx \frac{40}{2} (60 + 2(70 + 90 + 60) + 30)$ $= 10\,600 \text{ m}^2.$	x	0	40	80	120	160	y	60	70	90	60	30		y_0	y_1	y_2	y_3	y_4	① mark for applying Trapezoidal rule ① mark for correct answer
x	0	40	80	120	160															
y	60	70	90	60	30															
	y_0	y_1	y_2	y_3	y_4															
ii)	By joining the tips of the boundaries of the sections of the land, we form the trapeziums that the trapezoidal rule takes the sum of their areas.  It can be seen that the Trapezoidal Rule gives an area greater than the actual area (as the sum of the areas of the four trapeziums is greater than the actual area).	① mark for correct explanation																		

About 90% of students attained marks over 80%.

Question 13(15 marks) Use a SEPARATE writing booklet

(a)	Markers Comments
Find the values of a, b c for which $a + b(x + 2) + cx(x + 2) = 6x^2 + x - 2$ $LHS = a + bx + 2b + cx^2 + 2cx$ $= cx^2 + x(b + 2c) + a + 2b \quad \text{1 mark simplifying}$ $LHS = RHS$ $\therefore c = 6$ $b + 2c = 1 \quad \text{1 mark equating coefficients}$ $b + 12 = 1$ $\therefore b = -11$ $a + 2b = -2$ $a - 22 = -2 \quad \text{1 mark correct value a, b, c}$ $a = 20$ Alternative equate coefficients to find c=6 Put x=-2 then $a = 6(-2)^2 + (-2) - 2$ $a = 20$ put x=-1 both sides solve for b $20 + b(3) + 6(3) = 6 + 1 - 2$	Markers Comments Mostly done well if students remembered to equate coefficients some errors simplifying
(b)  In the diagram above, WXYZ is a rectangle with WX=12cm and WZ=9cm. WP is perpendicular to XZ. Copy the diagram onto your worksheet (i) Find the length of XZ (ii) Prove that $\triangle WXP$ is similar to $\triangle ZXW$. (iii) Hence find the length of XP. (i)	

In XYZ

$$XZ = \sqrt{XY^2 + YZ^2}$$

$$= \sqrt{9^2 + 12^2}$$

$$= 15 \text{ cm}$$

1 mark correct answer

In WXP and ZXW

$$\angle WPX = 90^\circ (\text{given})$$

$$\angle ZWX = 90^\circ (\text{WXYP is rectangle given})$$

$$\angle WXP = \angle ZXW (\text{common angle})$$

$$\therefore WXP \parallel ZXW (\text{two corresponding angles equal})$$

} 1 mark 2
} 90° angle
with reason

1 mark conclusion
with reason

(ii)

(iv) In WXP and ZWX

$$\frac{XP}{WX} = \frac{XW}{XZ} (\text{corresponding sides in similar s in same ratio})$$

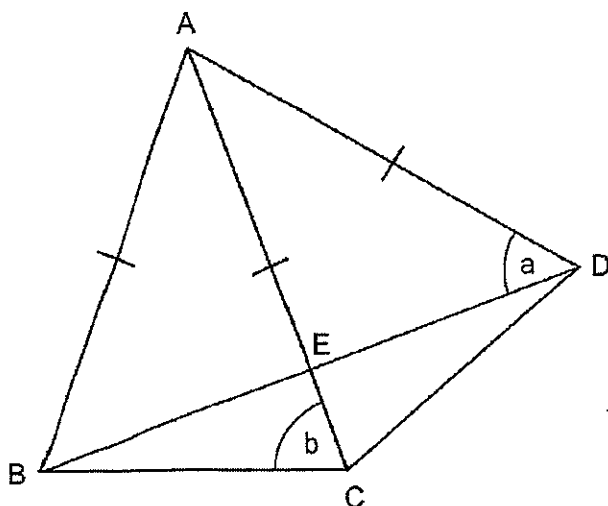
$$XP = \frac{12}{15} \times 12$$

1 mark correct length substituted with
reason

$$\therefore XP = \frac{144}{15}$$

ABC and ACD are congruent isosceles triangles.

Prove that $2b = a + 90^\circ$



Done well

An easy question if you
redraw the triangles
separately

Must state that

$\angle ZWX = 90$ because a
rectangle

Must use 3 letters to name
angles, $\angle A$ is not acceptable.

Using AA or AAA is not
accepted reasoning for
Similarity

Diagonals in rectangles do
not form 45° angles

A reason must be given for
1 mark

Some very long inefficient
solutions. Reasoning was
done poorly. Marks were
not given for working
without reasoning

- no 90° angle is given in
the question it must be
proved.

- ADCB is not given as a
kite it must be proved

- Base angles isosceles
triangle is not accepted
use "equal angles are
opposite equal sides"

In $\triangle ABC$ and $\triangle ACD$ $\angle ABC = \angle ADC = b$ (equal side opposite equal angles) <i>1 mark correct reasoning</i> $\angle ABD = \angle ABE = \angle ADE = a$ (equal sides opposite angles) <i>1 mark determining another angle</i> $\therefore \angle EBC = \angle EDC = b - a$ In $\triangle BDC$ $\angle EBC + \angle EDC + \angle DCB = 180^\circ$ (angle sum of a $\triangle$) $\therefore (b - a) + (b - a) + 2b = 180$ $4b - 2a = 180$ <i>1 mark final angle sum</i> $2b = a + 90$	- Corresponding angles in congruent triangles are the matching angles, not just the equal ones - Take care to name angles correctly
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(c) The graph of the parabola $(x - 3)^2 = 12(y - 2)$ is shown below. a) Find the coordinates of the vertex Vertex is (3,2) <i>1 mark</i> b) Find the coordinates of the focus and the equation of the directrix $(x - 3)^2 = 4(3)(y - 2)$ $\therefore a = 3$ focus at (3,5) <i>1 mark each focus and directrix</i> directrix at $y = -1$ c) Hence or otherwise show that the equation of the circle with centre at the focus passing through the vertex is: $x^2 - 6x + y^2 - 10y + 25 = 0$ $(x - 6x + 9) + (y^2 - 10y + 25) = 9$ <i>1 mark expanding and completing square</i> $(x - 3)^2 + (y - 5)^2 = 9$ this is a circle centre at (3,5) radius=3 <i>-1 mark for subsequent error</i> passes through (3,2) the vertex	Done well no working required can be solved by inspection Mostly done well Mostly done well
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Question 14

15 Marks

9) i) $P(\text{3 good starts}) = 0.85 \times 0.85 \times 0.85$
 $= 0.614125$

① answer.

ii) $1 - 0.85^n = 0.75$

① for first statement.

$$0.85^n = 0.25$$

$$n \ln(0.85) = \ln(0.25)$$

$$n = \frac{\ln(0.25)}{\ln(0.85)}$$

$$= 8.3048564$$

∴ Must start computer 9 times to have a greater than 75% chance of demonstrating the error.

b) i) $OA = OB = 0.6$ (radii of same circle)

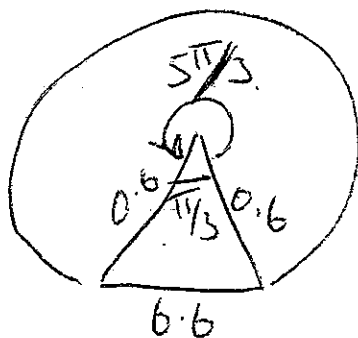
∴ $\triangle AOB$ equilateral as all sides = 0.6.

∴ all angles equal & $\pi/3$

∴ $\angle AOB = \pi/3$

① for reasoning
 $\triangle$ is equilateral

ii) 4



Area of table top = Area of sector + area of $\triangle$

$= \pi r^2 \times \frac{\theta}{2\pi} + \frac{1}{2}ab \sin C$

$= \pi \times (0.6)^2 \times \frac{5\pi/3}{2\pi} + \frac{1}{2}(0.6)^2 \sin(\pi/3)$

$= 1.098362364\pi$

$= 1.254246941$

$= 1.3 \text{ m}^2 \text{ (top)} \quad 1.1 \text{ m}^2$

Sector
 Segment

$\frac{1}{2}r^2\theta$
 $\frac{1}{2}r^2(\theta - \sin\theta)$

area of $\triangle$

13

$$\begin{aligned}
 c) \quad & 2 \int_0^{\pi/2} \sin 2x \, dx \\
 &= 2 \left[-\frac{\cos 2x}{2} \right]_0^{\pi/2} \\
 &= 2 \times -\frac{1}{2} \left[\cos 2x \right]_0^{\pi/2} \\
 &= - \left[\cos \pi - \cos 0 \right] \\
 &= - \left[-1 - 1 \right] \\
 &= -1 \times -2 \\
 &= \underline{2 \text{ units}^2}
 \end{aligned}$$

① for breaking up integral to 0 account for Neg. area.

① correct integration

① for final solution.

3

$$\begin{aligned}
 d) \quad i) \quad & 800 = 10e^{3k} \\
 & e^{3k} = 80 \\
 & 3k = \ln(80) \\
 & k = \frac{\ln(80)}{3} = 1.460675545
 \end{aligned}$$

① for substitution

① for answer.

2

$$ii) \quad 35000 = 10e^{kt}$$

① for substitution

$$3500 = e^{kt}$$

$$kt = \ln(3500)$$

$$t = \frac{\ln(3500)}{k}$$

$$= 5.586811031 \text{ yrs.}$$

$$\approx 5.6 \text{ yrs (1dp)}$$

① for answer.

2

iii) Limited availability of food resources

① for any valid reason.

1

Q14 $x = 3e^{-5t} + 3t - 2$

i) Initially: $t=0$

Displacement $x = 3e^0 + 3 \times 0 - 2$
 $= 1 \text{ m}$

Velocity $\dot{x} = -15e^{-5t} + 3$ ①

$t=0$ $\dot{x} = -15e^0 + 3$
 $= -12 \text{ m/s}$

Acceleration $\ddot{x} = 75e^{-5t}$
 $t=0$ $\ddot{x} = 75 \times e^0$
 $= 75 \text{ m/s}^2$

ii) Particle is stationary when $v=0$

$-15e^{-5t} + 3 = 0$

$e^{-5t} = \frac{1}{5}$, $-5t = \ln \frac{1}{5}$

$t = -\frac{1}{5}(\ln 1 - \ln 5) = \frac{1}{5} \ln 5 = 0.322 \text{ s}$

$x = 3e^{-5 \times \frac{1}{5} \ln 5} + 3\left(\frac{1}{5} \ln 5\right) - 2$

$= -0.434 \text{ m}$

$\ddot{x} = 75e^{-5 \times \frac{1}{5} \ln 5}$
 $= 15 \text{ m/sec}^2$

iii) as $t \rightarrow \infty$, $e^{-5t} \rightarrow 0$

$\therefore \dot{x} \rightarrow 0$, $\dot{x} = v \rightarrow 3 \text{ m/s}$

$x \rightarrow$ large positive value, that is
 it will continue moving in positive/
 forward direction.

3 marks correctly finds $x, \dot{x}, \ddot{x}$

2 marks correctly finds 2 of the
 $x, \dot{x}, \ddot{x}$.

1 mark evaluates 1 of the
 $x, \dot{x}, \ddot{x}$ (consider ecf)

3 marks correctly finds the
 $x, \dot{x}$ when $v=0$

2 marks correctly finds the
 time and $\dot{x}$ or $\ddot{x}$ OR

1 mark. Finds t when $v=0$ or evaluates
 x or $\ddot{x}$ (ecf).

2 marks. correct explanation

and shows $t \rightarrow \infty$

$e^{-5t} = \frac{1}{e^{5t}} \rightarrow 0$

1 mark any progress
 towards correct answer

$$14 b) \frac{dV}{dt} = 20 \sin \frac{\pi}{35} t$$

$$i) \frac{dV}{dt} = 0, 20 \sin \frac{\pi}{35} t = 0$$

$$\sin \frac{\pi}{35} t = 0, \pi$$

$$\frac{\pi}{35} t = \pi$$

$$t = 35 \text{ minute}$$

$\therefore$ First time after 8:55, $\frac{dV}{dt} = 0$ when

$$t = 8:55 + 35$$

$$= 9:30 \text{ pm}$$

$$ii) \frac{dV}{dt} = 20 \sin \left(\frac{\pi}{35} t \right)$$

$$V = \int 20 \sin \left(\frac{\pi}{35} t \right) dt$$

$$V = 20 \frac{-\cos \left(\frac{\pi}{35} t \right)}{\frac{\pi}{35}} + C$$

$$V = -\frac{700}{\pi} \cos \left(\frac{\pi}{35} t \right) + C$$

$$t=0, V=0 \quad \therefore 0 = -\frac{700}{\pi} \cos \left(\frac{\pi}{35} (0) \right) + C$$

$$C = \frac{700}{\pi} \times 1 \quad \therefore V = \frac{700}{\pi} - \frac{700}{\pi} \cos \frac{\pi}{35} t$$

iii) Filtering cycle is 35 min. (Part i)

$$\therefore V = \frac{700}{\pi} - \frac{700}{\pi} \cos \frac{\pi}{35} \cdot 35$$

$$= \frac{700}{\pi} - \frac{700}{\pi} \cos \pi$$

$$= \frac{700}{\pi} + \frac{700}{\pi}$$

$$= \frac{1400}{\pi} \text{ L}$$

2 marks correctly finds $t = 35 \text{ min.}$

1 mark any progress towards solution

2 marks Correctly finds the expression for 'Volume'.

1 mark Correct 'integration' to find 'volume'

3 marks Correctly evaluate the maximum value

2 marks correctly substitute $t = 35$ into formula and some progress towards solution

1 mark Any progress towards solution

Question 16

(a)

$$(i) \text{ Cost} = (12 \times 600) + (29 \times 450) \\ = \$20,250$$

1 for answer.
(with working)

(ii)

$$AB^2 = 600^2 + 450^2$$

$$AB = 750 \text{ m.}$$

$$\text{Cost} = 29 \times 750 \\ = \$21,750$$

1 for answer
(with working)

(iii)

$$DB = \sqrt{y^2 + 450^2} \quad AD = 600 - y$$

$$\text{Cost} = ~~29~~ 12 \times (600 - y) + 29 \times \sqrt{y^2 + 450^2}$$

$$\text{ie } F = 12(600 - y) + 29\sqrt{y^2 + 450^2}$$

1 for DB =
1 for AD =

$$(iv) F = 12(600 - y) + 29(y^2 + 450^2)^{\frac{1}{2}}$$

$$\frac{dF}{dy} = -12 + \frac{29}{2}(y^2 + 450^2)^{-\frac{1}{2}} \times 2y \\ = -12 + 29y(y^2 + 450^2)^{-\frac{1}{2}}$$

1 for working
towards

$$\frac{dF}{dy} = -12 + 29y(y^2 + 450^2)^{-\frac{1}{2}}$$

$$\frac{dF}{dy} = 0 \text{ when } \frac{29y}{(y^2 + 450^2)^{\frac{1}{2}}} = 12$$

$$\frac{841y^2}{y^2 + 450^2} = 144$$

$$841y^2 = 144y^2 + 29,160,000$$

$$697y^2 = 29,160,000$$

$$y^2 = \frac{29,160,000}{697} = 41836.44$$

$$y = 204.5 \text{ m}$$

1 for working
towards

$$y = 204.5 \text{ m}$$

Testing $y = 205.4$

y	200	204.5	210
$\frac{dF}{dy}$	-0.22	0	0.26

$\therefore$ Minimum Cost occurs when
 $y = 204.5$

$$\text{Cost} = F = 12 \times (600 - 205) + 29 \times \sqrt{205^2 + 450^2} \\ = \$19,080.34$$

v) F becomes

$$F = 12(600 - y) + 15(y^2 + 450^2)^{\frac{1}{2}} \quad \text{--- (1)}$$

$$\therefore \frac{dF}{dy} = -12 + \frac{15}{2}(y^2 + 450^2)^{-\frac{1}{2}} \times 2y \\ = -12 + 15y(y^2 + 450^2)^{-\frac{1}{2}}$$

$$\frac{dF}{dy} = 0 \text{ when } \frac{15y}{(y^2 + 450^2)^{\frac{1}{2}}} = 12$$

$$225y^2 = 144(y^2 + 450^2)$$

$$81y^2 = 29160000$$

$$y^2 = 360,000$$

$$\text{ie } y = 600 \text{ m}$$

$\therefore$ He will run the cable directly

from A to B ($y = 600$ means $AC = 0$ or $AD = 0$)

$$\text{Minimum Cost} = 15 \times (600^2 + 450^2)^{\frac{1}{2}} \\ (\text{from (1)}) = \$11,250$$

1 for showing
 $y = 204.5$ is
a Min Value

1 for cost

It is not
enough to show
A to B is the
cheapest of 3
scenarios.

Must show
working that
leads to AB
being cheapest
out of all
possibilities

1 $y = 600$

1 for saying
and showing A
is the Min path

(b)

(i) Max Value of $\sin(0.52x + 1.5) = 1$
 $\therefore \text{Max } T = 7 \times 1 + 13$
 $= 20^\circ\text{C}$

(ii) Min Value of $\sin(0.52x + 1.5) = -1$
 $\therefore \text{Min } T = 7 \times -1 + 13$
 $= 6^\circ\text{C}$

(iii) $T = 7 \sin(0.52x + 1.5) + 13$
Min Temp is 6°

$\therefore 6 = 7 \sin(0.52x + 1.5) + 13$

$-\frac{7}{7} = \sin(0.52x + 1.5)$

$0.52x = \sin^{-1}(-1) - 1.5$

~~$\sin^{-1}(-1) =$~~

$x = \left(\frac{3\pi}{2} - 1.5\right) \div 0.52$

$x \doteq 6$

$\therefore$ June has the lowest average temp

(iii) Find when the hottest month in Canberra is ie when $T = 20$ and note that this is ~~the~~ also the coldest month in New York

$-A \sin(0.52x + 1.5) + 13$

No marks
for no
working.

No marks for
wrong working
but ~~marks~~
answers correct

$\frac{1}{20} = \frac{1}{20}$

i If done
via a table
1 out of 2.

ii If read from
table 1 out of 2

In i
1 for working
towards

$T = 20$
1 for working
towards

ii $T = 6^\circ$
1 for showing

$6 = 7 \sin(0.52x + 1.5)$
1 for showing

solving to $x = 6$
and stating June
has lowest av

temp
iii 1 for showing
stating - Asen()

$$i) \frac{d}{dx} T = 7 \sin(0.52x + 1.5) + 13$$

$$T' = 7 \times 0.52 \cos(0.52x + 1.5)$$

$$T' = 3.64 \cos(0.52x + 1.5)$$

$$T'' = -3.64 \times 0.52 \sin(0.52x + 1.5) \\ = -1.8928 \sin(0.52x + 1.5)$$

$$T' = 0 \text{ when } \cos(0.52x + 1.5) = \frac{\pi}{2}, \frac{3\pi}{2}$$

$$0.52x = \frac{\pi}{2} - 1.5, \quad 0.52x = \frac{3\pi}{2} - 1.5$$

$$x = \frac{\frac{\pi}{2} - 1.5}{0.52} \\ = 0.1361$$

$$x = \frac{\frac{3\pi}{2} - 1.5}{0.52}$$

$$x = 6.1777$$

$$\text{when } x = 0.1361$$

$$\text{when } x = 6.1777$$

$$T'' = -1.89$$

$$T'' = 1.8928$$

$\therefore$ Max Value

$\therefore$ Min Value

$$\text{when } x = 6.1777$$

$$x = 0.1361$$

$$T = 7 \sin(0.52 \times 6.1777 + 1.5) + 13$$

when

$$\therefore T = 7 \sin(0.52 \times 0.1361 + 1.5) + 13 = 20$$

$$= 20$$

is the Max T

ii) From part i $x = 6.1777$ (Min Value via T'')

gives $T = 6$ (Min Temp)

$\therefore$ June gives the lowest.

Alternative
answer
for b i ii

1 for both
 x' values are
showing they
give Max +
Min Values
via T'' or
table.

1 for
answers
(must be
supported)

1 Must
have shown
that $x = 6.1777$
does give a
Min Value
1 for answer
(only if it can
be supported)